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Acceptability and Usability of Smileyscope Virtual Reality for Paediatric Pain Management During Burn Procedures: Perspectives of Patients, Carers and Clinicians

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ABSTRACT

Aim: To explore clinician, child and parent acceptability and usability of the Smileyscope VR device in the context of addressing the unique pain and distress needs of young burn patients.

Design: A survey comprising closed and open-ended questions.

Method: Descriptive statistics analysed participant characteristics, pain and analgesia. Qualitative content was collected from April 2022–August 2022 and analysed to identify barriers and enablers. Categories were then mapped onto the Capabilities, Opportunities and Motivation-Behaviour Wheel (COM-B) framework.

Results: Smileyscope was found to be effective for reducing pain and anxiety during dressing changes by both patients ($n=39$) and parents ($n=37$). Clinicians ($n=35$) reported high self-efficacy and willingness to reuse the device. However, concerns arose regarding the device's fit and the need for age-appropriate programmes.

Conclusion: Smileyscope demonstrated promise in reducing procedural pain and distress. The device is well accepted by all participants implying ease of implementation. Feedback suggests further program development and fitting optimisation is required.

Implications for the Profession and/or Patient Care: Improved procedural pain has proven to decrease wound healing times, decreasing possible need for further scar management and long-term consequences after sustaining a burn injury. Smileyscope use in rural hospitals presents valuable opportunities for optimising early paediatric burn pain.

Impact: Increased burn pain can delay wound healing and have long term physical and psychological impact on patients. Smileyscope was well received within this cohort; however, improvements in device design and programmes were suggested. This study shows the potential for use of Smileyscope as a non-pharmacological approach to improving paediatric burn pain and distress.

Patient or Public Contribution: While our study included patients, parents and clinicians as research participants, there was no patient or public contribution in the design or conduct of the study, analysis or interpretation of the data.

Summary

What Already Is Known

- Acute burn dressing changes are painful and distressing for paediatric patients.
- Combining pharmacological and non-pharmacological techniques is essential for effective pain management.
- VR has potential in reducing pain and anxiety.

What This Paper Adds

- Smileyscope effectively reduces pain and anxiety during dressing changes, with high satisfaction from patients and parents.
- Clinicians found it easy to use and are willing to continue, though improvements in fit and age-specific content are needed.
- Patients required minimal analgesia, with some needing no additional pain medication.

Implications for Practice/Policy

- Smileyscope shows promise as a non-pharmacological approach to improving paediatric burn pain and distress.
- Further development is needed to enhance the fit and programme content for diverse age groups.
- The device could be particularly useful in managing early paediatric burn pain in rural hospitals.

1 | Introduction

Clinicians continue to strive for effective techniques to alleviate pain and anxiety in paediatric patients undergoing clinical procedures, especially those grappling with painful conditions such as burn injuries (Holbert et al. 2021). The level of pain and anxiety reported by these young patients is contingent upon factors such as percentage of total body surface area burned, and depth of burns, along with previous prior medical experiences which can profoundly influence their overall wellbeing outcome (Holbert et al. 2021). Previous research firmly establishes that the relationship between pain and anxiety significantly influences the process of wound re-epithelialisation (Brown, Kimble, Gramotnev, et al. 2013; Miller et al. 2011). Furthermore, elevated levels of pain and anxiety can also contribute to long-term psychosocial implications for both paediatric patients and their parents or caregivers (Brown et al. 2014, 2018; De Young et al. 2014).

Paediatric patients are faced with a variety of situations, including the traumatic nature of the burn injury, the anticipation of impending painful procedures, previous experiences within the hospital setting, prolonged separation from family members, or a sense of powerlessness in their situation that can lead to excessive distress (Fagin and Palmieri 2017). The recovery journey often involves enduring multiple painful wound dressings and procedures, necessitating the use of opioids and anxiolytic sedating agents (Bayat, Ramaiah, and Bhananker 2010; Hansen et al. 2019). However, they are not always appropriate (Gillum et al. 2022) and can impact hospital length of stay and procedure recovery time (Hansen et al. 2019). Clinicians increasingly recognise the necessity of a comprehensive approach, extending beyond the conventional, stand-alone, pharmacological approach (Chu et al. 2021; Kaheni et al. 2016).

As a result, non-pharmacological approaches, such as music therapy, hypnotherapy, play therapy, the presence of clown doctors and child life specialists, and immersive virtual reality (VR) experiences, have been utilised to manage pain and anxiety in clinical procedures including paediatric burn dressings (Gillum et al. 2022; Hornsby, Blom, and Sengoelge 2020). These innovative approaches not only alleviate the distress experienced by young patients but also enhance the overall quality of care by addressing their emotional psychological needs (Balanyuk et al. 2018; Chu et al. 2021; Kaheni et al. 2016).

There is a growing body of literature reporting the use of VR in both adult and paediatric burns care with largely positive effects for reducing pain, distress and anxiety (Brown, Kimble, Rodger et al. 2013; Hoffman et al. 2000, 2001; Jeffs et al. 2014; Kipping et al. 2012; Patterson et al. 2022). The immersive effect of VR is thought to isolate the patient from what is occurring in the room and submerse them into a virtual computer-generated world (Hoffman et al. 2011). Some studies have shown up to a 30%–50% decrease in reported pain with the use of VR (Hoffman et al. 2011). Virtual reality is a unique alternative to conventional devices like phones and tablets, demonstrating its effectiveness in immersing patients and diverting their attention from painful procedures in diverse clinical contexts, including dental treatments, cancer therapies and burn wound management (Chan et al. 2019; Gold et al. 2021).

Studies assessing the effectiveness of VR during burns dressings often use different devices, yielding inconsistent results (He et al. 2022). For example, Kipping et al. tested an off-the-shelf VR device (Kipping et al. 2012) with nursing staff reporting a statistically significant reduction in pain scores during dressing removal, yet the self-reported pain scores for the VR group did not exhibit a statistically significant difference. While the results were still promising, it was suggested that this complex off-the-shelf VR device, containing a headset and joystick hand control, may not be as effective compared to more customised ones. Patterson et al and Hoffman et al addressed bulky VR limitations in a wet areas by using water-resistant devices consisting of a medical cart with an articulated arm that holds the VR goggles and a wireless mouse (Hoffman et al. 2019; Patterson et al. 2022). Despite significant pain reduction, both approaches required substantial equipment and time-consuming tech support. The accessibility and usability of innovative technology such as VR is a critical element of implementation for integration into real-world clinical practice (Zanatta et al. 2022).

The Smileyscope device (Smileyscope Holding Inc, Melbourne, Australia), offers a novel alternative by utilising a headset controlled by the patient's head movements and a smartphone interface. While it has been found to be acceptable and efficient in alleviating pain and anxiety during needle procedures (Chan et al. 2019), its potential utility in paediatric burns has not been explored. This study aims to evaluate the acceptability and usability of integrating Smileyscope into our paediatric burn's outpatient department by employing principles of implementation science.

2 | Methods

2.1 | Design

In this study, we used a survey approach to generate both quantitative and qualitative data. The survey was grounded in the principles of implementation science, with a specific focus on the acceptability and usability of integrating Smileyscope into the context of paediatric burn dressing changes (Bauer and Kirchner 2020). The Theoretical Framework of Acceptability (TFA) was used to construct the surveys with focus on four main constructs: affective attitude, burden, perceived effectiveness and self-efficacy.

Qualitative and quantitative data were systematically merged during the interpretation phase to identify both the enablers and barriers encountered in implementing Smileyscope, thus allowing for an exploration of its overall acceptability within paediatric burn care settings (Fetters, Curry, and Creswell 2013).

2.2 | Study Setting and Recruitment

This research was conducted at the Pegg Leditschke Children's Burns Centre, located within the Queensland Children's Hospital, Brisbane between April 2022 and August 2022. This centre serves as the primary referral centre for paediatric burns across Queensland, providing comprehensive care for young burn patients up to the age of 16 in the region.

The study encompassed three distinct participant groups: children, parents and carers and clinicians. Eligible participants among children ranged from 4 to 16 years of age, with exclusion criteria comprising non-English speaking children and those who had sustained facial burns. Verbal consent was sought from both parents/carers, as well as patients, to collect feedback on their experience with Smileyscope following their burns dressing procedure. Children and parents were under no obligation to complete the surveys (Appendix S1), nor did it impact their ability to access the device for their clinical procedure. The completion of surveys was undertaken by parent/caregivers, clinicians and children eight or older.

2.3 | Procedures

2.3.1 | Intervention

In conjunction with standard pain relief protocols, Smileyscope was used as an adjunctive intervention. Pain relief by oral medication(s) followed the established clinic procedures and allowed appropriate time to take effect. Additional pain medication was administered as deemed necessary based on the patient's clinical need.

Clinicians worked with children to select one of over 20 immersive Virtual Reality experiences. To prepare children for the clinical procedure, Smileyscope was introduced no <5 min prior to the clinical procedure commencing. This enabled the child to become immersed within the program and less distracted by what was occurring in the room. Following the completion of the VR therapy program, children had the option to explore other program within the device or repeat the same program until the clinical procedure was completed. Upon conclusion of the procedure, all stakeholders, including parents, patients and clinicians were invited to complete the survey, providing valuable feedback on their respective experiences and perspectives.

2.3.2 | Measures

2.3.2.1 | Patient Data. Demographic data including age, gender, mechanism of injury, burn depth and location of injury, was extracted from the electronic medical records of all participating children. Pain scores recorded by clinicians, as well as analgesia requirements, were also extracted from electronic medical records.

2.3.2.2 | Survey. Three custom designed survey**s** were developed for participants to complete: one for patients, one for parents and one for clinicians (Appendix S1) involved in the use of Smileyscope VR during burn dressing changes. The survey underwent multiple rounds of iterative refinement based on input from burns clinicians to enhance its clarity and ease of use. The survey primarily focused on stakeholders' engagement and their experiences of Smileyscope, using elements of the TFA (Sekhon, Cartwright, and Francis 2022) to assess the intervention's acceptability within the context of burns dressing changes. The survey comprised of age-appropriate pain scales, 5-point Likert scales and open-ended questions, encouraging participants to list barriers and enablers to their experience and use of the Smileyscope VR headset. Specifically, the survey focused on the following components.

2.3.2.3 | Acceptability and Usability. These surveys were aligned with the TFA constructs of affective attitude, burden and perceived effectiveness, (Sekhon, Cartwright, and Francis 2022). Following consultations with key stakeholders, questions were adapted to reflect the environment and age of the patients (Appendix S1). Participants were asked to rate: their overall experience on the device, the importance of access to the device and to provide recommendations for its future use in procedures. Additionally, Table 1 illustrates the TFA constructs used and example questions from the surveys.

2.3.2.4 | Pain and Distress. Pain scores were assessed by clinicians at three distinct time points: pre-procedure, peak procedure (during cleaning of wound) and post procedure using the validated Faces, Legs, Activity, Cry, Consolability (FLACC) pain scale (Crellin et al. 2021). Children were asked to rate their pain using the Faces Pain Scale Revised (FPS-R) (Hicks et al. 2001), before and during their procedure. While the FPS-R has been validated for children as young as four and is recommended for research on this age group based on its psychometric properties (Tomlinson et al. 2010), a deliberate decision was made to limit the inclusion of participants to those aged eight or older in the surveys. This choice was also influenced by the incorporation of other scaled and open-ended questions in the survey that demand a higher level of comprehension. Subsequent to the procedure, parents were also invited to rate

TABLE 1 | TFA constructs and example questions.

Construct	Definition	Participant	Example question
Affective attitude	How did the individual feel about the intervention	Patient	Did you enjoy the videos?
Burden	Amount of effort required to participant	Patient	Were the goggles difficult to wear?
Perceived effectiveness	How effective was the intervention	Patient	Do you think the goggles helped?
—	—	Parent	Do you think the Smileyscope helped your child's experience?
—	—	Clinician	How helpful was the Smileyscope during the procedure?
—	—	Clinician	Do you think Smileyscope may improve pain and anxiety scores?
Self-efficacy	Confidence that they can use the intervention	Clinician	How likely are you to use the Smileyscope in future procedures?

their child's worst pain during the dressing change using a 11-point numeric rating scale (Lifland et al. 2018).

Analgesic medication use was recorded in the electronic medical records for all children, with the clinic primarily prescribing a standard combination or selection of paracetamol, ibuprofen or oxycodone. In cases necessitating additional analgesia, intranasal fentanyl was administered. The selection of medications considered numerous factors, including the risk of increased pain response, as identified by Holbert et al. (2021). Additionally, parents rated their level of anxiety about their child's procedure using a 11-point numeric rating scale, acknowledging the documented impact of parental anxiety on a child's coping and wound healing in previous studies (Brown et al. 2018, 2019a).

2.4 | Data Analysis

Simple descriptive analysis was used to summarise the characteristics of the children (including age, gender, first aid history, burn injury severity) experiencing the Smileyscope using frequency distributions, means, standard deviations and ranges. Descriptive findings are organised into tables, each focusing on specific aspects of the Smileyscope experience.

Pain and analgesia data were analysed using frequency distributions to illustrate the experience of pain levels among the participants. Additionally, means and standard deviations were utilised to present variability within the sample and provided additional insight into the dispersion of pain experiences among the children.

An inductive content analysis approach, following the methodology outlined by Vears and Gillam (He et al. 2022), was used for qualitative analysis. This approach is suitable for analysing qualitative date derived from open-ended research questions in surveys. This involved creating an inductive summary of multiple individual texts (Vears and Gillam 2022), starting with gaining familiarity with the free-text responses. Initial coding identified overarching meaning units such as 'Age appropriateness' and 'Reduced pain and anxiety'. Subsequent phases involved a detailed examination resulting in the identification of barriers, enablers and specific codes for statement categorisation.

This iterative coding process allowed for refining, generating and consolidating categories and codes. The final phase included synthesis and interpretation, mapping onto the Capabilities, Opportunities and Motivation-Behaviour (COM-B) Wheel theoretical framework (Michie, van Stralen, and West 2011). This comprehensive qualitative analysis aimed to uncover perceived barriers and enablers among patients, carers and clinicians providing valuable insights into the Smileyscope VR adoption and utilisation. Integration with quantitative measures enhances the understanding of the factors influencing the usage of Smileyscope VR.

2.5 | Ethics Statement

This quality assurance project was reviewed and approved by the Children's Health Queensland Research Ethics Committee (CHQ HREC) (EX/22/QCHQ/85458). Verbal consent was obtained from all children and their parents or caregivers and clinicians.

3 | Results

In total, $n = 39$ children, $n = 37$ parents and $n = 35$ clinicians participated in this study. There was some variability in the surveys due to missing data; therefore, each individual question had different n totals.

3.1 | Demographics

Table 2 illustrates the patient demographics. Of this cohort, 54% were male, with a mean age of 7.4 years (Standard Deviation [SD]: 3.05 years). The mean total body surface area (TBSA%) affected by burns was 1.7% (1.8). Contact burns and scald burns accounted for over 50% of the mechanisms of injury, with superficial partial thickness burns being the most common depth of

TABLE 2 | Patient demographics.

	n	Percent	Mean (SD)	Min–Max
Age (years)	39	—	7.4 (3.05)	3–14
Gender				
Male	21	54%		
Female	15	38%		
Not specified	3	8%		
TBSA (%)	39		1.7 (1.8)	0.5–8
Mechanism of injury				
Contact	14	36%		
Scald	13	33%		
Friction	5	13%		
Electrical	1	3%		
Extravasation	1	3%		
Not specified	5	13%		
Depth (at deepest)				
Superficial partial thickness	26	67%		
Deep partial thickness	6	15%		
Full thickness	2	5%		
Not specified	5	13%		
Burn locations				
1 Location	20	51%		
2 Locations	11	28%		
3 Locations	3	8%		
Not specified	5	13%		
First presentation				
QCH	17	43.5%		
Other	17	43.5%		
Not documented	5	13%		
First aid (Cold running water [CRW])				
≥20 min	22	56.4%		
<20 min	6	15.3%		
None	6	15.3%		
Not documented	5	13%		
Surgical debridement in operating room <24 h post injury				
Yes	3	7.6%		
No	31	79.4%		
Not documented	5	13%		

injury. In this study, however, no demographic information was obtained from parents or clinicians.

3.2 | Quantitative Results

3.2.1 | Pain Scores

Pain scores were collected from self-reporting patients ($n=15$) and clinicians ($n=34$), along with self-reported anxiety scales from the parents ($n=32$) are summarised in Table 3. The mean peak pain ratings for clinicians and patients were 1.7 (SD: 0.3) and 3.7 (SD: 2.6), respectively.

3.2.2 | Analgesia

Analgesic medication use for all children ($n=39$) is summarised in Table 4. Seven children did not require any analgesia or were not specified. Intranasal opioids were required in two children. Sedation was not required for any patient in this study.

TABLE 3 | Clinician, patient and parent reported pain scores.

	Mean (SD)	Min–Max
Clinicians[a] ($n=33$)		
Pre	0.1 (0.1)	0–3
Peak	1.7 (0.3)	0–4
Post	0 (0)	0–0
Patient[b] ($n=15$)		
Pre	3.7 (2.6)	0–10
During	4 (3.2)	0–10
Parent/Carer[c] ($n=32$)		
During	4.5 (2.3)	0–10

[a]Faces, Legs, Arms, Crying and Consolability (FLACC).
[b]Faces Pain Scale – Revised (FPS-R).
[c]Numeric Rating Scale (NRS).

TABLE 4 | Analgesia used.

	$n=39$	Percent
Ibuprofen	1	3
Oxycodone	1	3
IN fentanyl	1	3
Oxycodone, paracetamol, ibuprofen	22	56
Oxycodone, paracetamol, ibuprofen, IN fentanyl	1	3
Oxycodone, paracetamol	2	5
Oxycodone, ibuprofen	1	3
Paracetamol, ibuprofen	3	8
Nil/Not specified	7	18

3.2.3 | Acceptability Constructs

The results of the clinician, patient and parent reported acceptability constructs have been tabulated and reported in Table 5.

3.2.3.1 | Affective Attitude. Patients attitude was measured in terms of how much they liked or disliked the Smileyscope videos. A total of 60% ($n=9/15$) of patients stated they liked the videos either 'a lot' or 'quite a lot'.

3.2.3.2 | Burden. Assessment of burden focused on the fit of the Smileyscope device. 47% ($n=7/15$) of patients felt the goggles were 'somewhat' heavy or difficult to wear. While 33% ($n=5/15$) experienced them as 'a bit' heavy or difficult to wear. Only 20% ($n=3/15$) of patients reported no concerns with the fit of the goggles.

3.2.3.3 | Perceived Effectiveness. We were able to assess perceived effectiveness in all three groups. All patients felt that Smileyscope assisted them during their procedure, with 77% ($n=11/15$) of patients stating that Smileyscope helped them during their procedure either 'a bit' or 'quite a lot'. Parents unanimously, 100% ($n=37/37$), felt that the device helped their child throughout their procedure, either 'a lot' or 'a bit'.

Clinicians felt the Smileyscope 'very helpful' or 'helpful' in assisting with the procedure, 86% ($n=30/35$). Clinicians also perceived effectiveness that Smileyscope may have improved pain and anxiety, with 87% ($n=26/30$) either agreeing or strongly agreeing to this. However, 13% ($n=4/30$) felt that using Smileyscope did not improve pain and anxiety during the procedure.

3.2.3.4 | Self-Efficacy. Considering the Integration of Smileyscope Into the Unit, all Clinicians That Completed This Question, $n=32$, Expressed Their Willingness to Use the Device Again for Clinical Procedures, Indicating a High Level of Confidence in Its Effective Implementation.

3.3 | Qualitative Results

Results of the content thematic analysis, combined with the inductive coding guided by the Capabilities, Opportunities and Motivation-Behaviour (COM-B) Wheel theoretical framework, identified perspectives of patients, parents and clinicians regarding their experiences with Smileyscope VR, as summarised in Table 6.

3.3.1 | Capability

Reduction in pain was the leading enabler of using the Smileyscope and was mentioned by all three reporting groups. Patients self-reported several enabling aspects, including finding the experience enjoyable, benefiting from distraction

TABLE 5 | Clinician, patient and parent reported acceptability constructs.

		n (%)					Median (Interquartile range)
Construct	**Item (participant group)**	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	
Affective attitude	Did you enjoy the videos? (Patient, $n=15$)	Not at all 0 (0)	Yes, a bit 5 (33)	Somewhat 1 (7)	Yes, quite a bit 3 (20)	Yes, a lot 6 (40)	4 (3)
	How important do you think it is to have Smileyscope VR at every dressing change? (carer, $n=27$)	Yes, a lot 21 (78)	Yes, a bit 5 (19)	Maybe 1 (4)	Not really 0 (0)	Not at all 0 (0)	1 (0)
Burden	Were the goggles heavy or difficult to wear? (Patient, $n=15$)	Not at all 3 (20)	Yes, a bit 5 (33)	Somewhat 7 (47)	Yes, quite a bit 0 (0)	Yes, a lot 0 (0)	2 (1)
Perceived effectiveness	Do you think the goggles helped your procedure today? (Patient, $n=15$)	Not at all 0 (0)	Yes, a bit 2 (13)	Somewhat 2 (13)	Yes, quite a bit 6 (40)	Yes, a lot 5 (33)	4 (2)
	How much did your dressing change bother you? (Patient, $n=15$)	Not at all 1 (7)	Yes, a bit 9 (60)	Somewhat 5 (33)	Yes, quite a bit 0 (0)	Yes, a lot 0 (0)	2 (1)
	Do you think the Smileyscope VR helped your child's experience today? (Carer $n=37$)	Yes, a lot 30 (81)	Yes, a bit 7 (19)	Maybe 0 (0)	Not really 0 (0)	Not at all 0 (0)	1 (0)
	How helpful was Smileyscope VR during the procedure? (Clinician, $n=35$)	Very helpful 20 (57)	Helpful 10 (29)	Neutral 4 (11)	Not helpful 1 (3)		1 (1)
	Do you think Smileyscope VR may improve pain and anxiety scores? (Clinician, $n=30$)	Strongly Disagree 4 (13)	Disagree 0 (0)	No opinion 0 (0)	Agree 11 (37)	Strongly Agree 15 (50)	4.5 (1)
Self-efficacy	How likely are you to use Smileyscope VR for future procedures? (Clinician, $n=32$)	Not at all 0 (0)	Unlikely 0 (0)	Neutral 0 (0)	Likely 4 (13)	Very Likely 28 (88)	5 (0)

TABLE 6 | Overview of analysis showing barriers and enablers mapped to the COM-B framework Domains.

COM-B Domain	Enabler/Barrier	Category	Relevant participant group
Physical capability	Enabler	Reduced pain and anxiety	Clinician, parent, patient
	Barrier	Still bothered by dressing change	Patient
	Enabler	Easier to do dressing	Clinician, parent
	Enabler	Enjoyable	Patient
Psychological capability	Barrier	Age appropriateness and variety of programmes	Clinician, parent patient
	Enabler	Induced relaxation	Patient
Physical opportunity	Barrier	Availability of devices	Clinician
	Barrier	Patient position	Clinician
	Barrier	Logistical issues with the device	Parent, patient
	Enabler	Less resources used	Clinician, parent
Automatic motivation	Enabler	Enjoyable and distracting	Clinician, parent, patient

during procedures, experiencing reducing pain sensations and inducing relaxation. One patient described how the burns dressing was 'not that painful when I was watching the VR' (Patient 19, aged 14). Parents and clinicians agreed and highlighted the positive effects of Smileyscope VR, underscoring its role in distraction, anxiety reduction and pain relief for their children during procedures. A clinician shared their appreciation of the impact of VR, stating they were 'able to debride wound with nil issues' (Clinician of patient 19, aged 14). Parents also highlighted how VR minimised trauma during the burns dressing, as opposed to a previous attempt in the emergency department:

> They made what I thought would be a distressing event, stress free. All dead skin was effectively removed with minimal trauma. They attempted same in the ED the night before with pain meds only, he carried on like a pork chop, and we had to redo it the next day because they weren't successful.
>
> (Parent of patient 38, age 8)

Procedural anxiety was a common concern of both patients and parents, but qualitative feedback highlighted the Smileyscope VR's effectiveness in providing distraction. It transformed an often-distressing procedure into a stress-free experience, allowing the patient to concentrate on coping strategies without requiring pain relief. A clinician highlighted this, noting their 'patient was able to concentrate on breathing throughout dressing. No oral pain relief required' (Clinician of patient 39, age 8). Parents echoed this sentiment, expressing appreciation for the 'immersive and distracting' nature of Smileyscope, appreciating the sound feature of the Smileyscope 'so parents could follow and encourage' (Parent of patient 16, aged 6). Another parent highlighted the positive impact of distraction for their child, stating, 'the Smileyscope was a great distraction for (Child). (Child's)'s anxiety reduced by the device greatly...' (Parent of patient 13, aged 4).

The reoccurring barrier to the use of Smileyscope within this cohort focused on the programmes available for the patients. One clinician suggested they were not appropriate or engaging for some ages and recommend that the programmes were more 'targeted at a younger age than teenager, would suggest an older program' (Clinician of patient 21, aged 14). A patient expressed dissatisfaction with the program used, stating, 'boring' (Patient 7, aged 6). One parent praised the concept but found room for improvement, stating, '(Child) said the video was a bit boring. But the idea is great' (Parent of patient 39, aged 8). Patient, parent and clinician comments suggested that the programmes available may require further review to ensure they cater for a wider variety of age groups, suggesting potential enhancements. Another parent commented, 'I think in time there could be different things to watch according to age' (Parent of patient 12, aged 8).

3.3.2 | Opportunity

The Smileyscope was noted by all groups to improve patient outcomes with minimal other resources required. A clinician mentioned they 'felt I could have done the procedure with fewer resources (e.g., did not need another staff member, used less pain relief)' (Clinician of patient 24, age unknown), suggesting that VR technology such as Smileyscope can enhance patient experiences, reduce resource utilisation and optimise procedural efficiency. Parents also welcomed the use of the device as an adjunct to analgesia or instead of analgesia and acknowledged the need for devices in this setting:

> It's an amazing support tool for anxious children and takes them momentarily to another place. Any distraction to assist is amazing.
>
> (Parent of patient 2, age 6)

However, device availability impacted clinician's ability to provide this resource to patients. An increase in devices available enabled clinicians to utilise them more effectively:

> Now we have 2 (devices), (we are) able to use more sparingly. Previously when (we) only had 1, (we) found it hard if (the device was) being already used.
>
> (Clinician of patient 4, age 8)

> Parents also acknowledged the need for the devices stating, 'Please provide the Burns Unit with VR's as it makes horrible experiences less painful to the little kids.
>
> (Parent of pt 11, aged 8)

3.3.3 | Motivation

The use of Smileyscope stemmed from clinicians need to improve patient outcomes and recognising the importance of taking a multimodal approach to pain and anxiety. A parent stated, 'Anything that can help to child [sic] and the parents/carers to make the situation a more positive experience is a must' (Parent of patient 2, age 6). All groups acknowledge that the use of Smileyscope within this clinical setting improved patient experience while providing distraction for notably painful procedures. According to a patient, the experience was 'fun and enjoyable, funny characters,' and 'it distracted me, it was like kittens are scratching you. The fish were my favourite' (Patient 5, aged 9). A clinician noted, 'Patient was calmer and more cooperative, parents/carers were less anxious and/or better able to support, saved time' (Clinician of patient 22, aged 9).

3.4 | Integration of Qualitative and Quantitative Results

In this study, the integration of qualitative and quantitative survey data served as an important aspect of the research methodology. Throughout the interpretation phase, researchers sought points of convergence between these data types, ensuring a comprehensive understanding. A cross-case comparison joint display (Table 7) (Guetterman, Fetters, and Creswell 2015) was employed to visualise the integration process, facilitating an exploration of the combined findings. For instance, while quantitative metrics, such as pain scores and acceptability scores, provided numerical insights into the efficacy of Smileyscope, qualitative analysis revealed specific barriers and enablers of acceptability and behaviour. By analysing and integrating this data, researchers were able to offer a multifaceted interpretation of the survey data, enhancing the study's insights and implications for clinical practice.

4 | Discussion

There is documented evidence that pain management within acute burns dressings continues to elude clinicians and inadequate pain management can lead to a decrease in wound healing resulting in long-term psychosocial effects on both patients and families (Brown et al. 2014; De Young et al. 2012; Summer et al. 2007). This study is the first to examine the acceptability and usability of a VR device, within the paediatric burn's environment. This study demonstrated that minimal analgesia was required for our patients with relatively low reported pain scores. Patients felt that the use of the VR device helped them throughout their procedure and the majority of patients liked the current available videos. Findings from the qualitative analysis demonstrated that the use of the VR device appeared to reduce the experience of pain, anxiety and stress for both patients and parent. Suggestions were made that further development of more age specific programmes and further review of the fit of the goggles would be beneficial. These findings provide directions for how Smileyscope VR can be adapted and integrated during paediatric burn wound care.

4.1 | Reduction in Pain

Pain can be influenced by anxiety and stress as well as physical aspects of a burn injury including depth, mechanism, location of burn and total body surface area affected (De Young et al. 2012; Holbert et al. 2021). Holbert et al. (2021) showed particular areas of the body were more susceptible for increasing pain including hands and feet, multiple anatomical regions burnt and initial treatment in a non-burn centre. From our cohort of patients, 33% of patients sustained burns to hands and feet, 49% sustained burns to multiple areas while 43.4% had their initial treatment at a non-burns centre. This highlights the increased need to ensure adequate pain management processes are implemented.

There has also been a push in recent times to take a more holistic approach to pain management with the focus on combining both pharmacological and non-pharmacological techniques for improved outcomes (Griggs et al. 2017; Preston and Ambardekar 2020; Srouji, Ratnapalan, and Schneeweiss 2010). Opioid use in burns care seems to be standard treatment; however, patients often require further procedural pain management during dressing changes and wound debridement. Reducing opioid use is crucial in managing long term negative effects from medication use (Holtman Jr. and Jellish 2012; Kim et al. 2019). Standard pain management care within our facility consists of a combination or paracetamol, ibuprofen and oxycodone. Further procedural pain management can be given in the forms of intranasal fentanyl, Entonox and intranasal midazolam, all of which are prescribed using our current hospital guidelines for procedural pain management within ambulatory areas. Only two patients within this cohort required the use of intranasal fentanyl for procedural pain management in combination with the use of Smileyscope. One patient was prescribed fentanyl due to their inability to take oral medication. This in combination with Smileyscope was the only medication used in this instance. The combination of pharmacological and non-pharmacological techniques had a noticeable affect for some patients within our study. It was also noted by parent and clinician that in some cases, the sole use of Smileyscope provided enough distraction that no medication was required.

Past medical experiences, trauma and family influence have also shown to play a large part in a child's pain and anxiety (Brown et al. 2019b). Children are influenced by their parents in unfamiliar settings, and factors including parental guilt, anxiety and PTSS can increase child's anxiety (Brown et al. 2019a). Studies have shown the importance of clinicians building rapport with parents to improve their ability to effectively communicate

TABLE 7 | A cross-case comparison joint display of the data, showing integrated qualitative and qualitative findings.

Domain	Variables	Clinician	Parent/Carer	Patient	Integrated findings
Reduction in pain and distress	Pain—peak scores, Mean (SD)	1.7 (0.3)	4.5 (2.3)	4 (3.2)	• Patient and carer pain scores congruent; clinician-reported pain scores lower • High perceived effectiveness for reducing pain/anxiety • Clinicians and parents reported successful procedures without additional analgesia/resources, also shown in clinician reported analgesia • Some patients still bothered by dressing changes, indicating individual variances in pain reduction efficacy
	Analgesia	Clinician reported, 7 children did not require analgesia/not specified. 2 required intranasal opioids.			
	Perceived effectiveness	Do you think Smileyscope VR may improve pain and anxiety scores? Median: 4.5, IQR: 1	Do you think the Smileyscope VR helped your child's experience today? Median: 1, IQR: 0	How much did your dressing change bother you? Median: 2, IQR: 1	
	Enablers-Reduced pain, less resources used Barriers-Still bothered by dressing change	'(patient name) was distracted while I was cleaning the burn, she said it was helping her' (Clinician of pt 6, age 14)	'(patient name) was totally distracted with the VR. It definitely made the experience less painful' (Parent of pt 11, age 8)	'Kept me distracted and I didn't feel much pain at all' (Patient 34, age 8). 'It stung when they washed the burn' (Patient 6, age 14)	
Programmes and device	Burden			Were the goggles heavy or difficult to wear? Median: 2, IQR: 1	• Moderate burden for goggles • Concerns about age appropriateness, program variety and logistics noted in qualitative data • Suggestions for more targeted programs to improve acceptability • Clinicians highlighted the need for more devices
	Barriers – Age, programmes, availability, logistical issues	'More interactive programs in procedure section would be more beneficial' (Clinician of pt 11, age 8)	'Better head straps, my daughters head was too small for the headset' (Parent of pt 5, age 9). '(Patient name) said the video was a bit boring. But the idea is great' (Parent of pt 39, age 8)	'Would like to watch a different one next time' (Patient 32, age 7) 'The bottom digging into top of cheek. Would to be [sic] made aware of the options' (Patient 22, age 9)	

(Continues)

TABLE 7 | (Continued)

Domain	Variables	Clinician	Parent/Carer	Patient	Integrated findings
Compliance	Perceived effectiveness	How helpful was Smileyscope VR during the procedure? Median: 1, IQR: 1	Do you think the Smileyscope VR helped your child's experience today? Median: 1, IQR: 0	Do you think the goggles helped your procedure today? Median: 4, IQR: 2	• All groups found helpful, high perceived effectiveness enhancing compliance. • Patients enjoyed the distraction • Clinicians and parents noted that Smileyscopemade dressing changes easier
	Affective attitude		How important do you think it is to have Smileyscope VR at every dressing change? Median: 1, IQR: 0	Did you enjoy the videos? Median: 4, IQR: 3	
	Enablers-Easier to do dressing, less resources, enjoyable and distracting Barriers – Age, need for control	'Patient was calmer and more cooperative, parents/carers were less anxious and/or better able to support, saved time' (Clinician of pt 22, age 9) 'Age, not appropriate for toddlers' (Clinician of pt 13, age 4)	'They made what I thought would be a distressing event, stress free. All dead skin was effectively removed with minimal trauma. They attempted same in the ED the night before with pain meds only, he carried on like a pork chop and we had to redo it the next day because they weren't successful' (Parent of pt 38, age 7) 'More so for younger children (ours is 14) it would provide a valuable distraction' (Parent of pt 31, age 14)	'Fun and enjoyable, funny characters, "It distracted me, it was like kittens are scratching you. The fish were my favourite"' (Patient 5, age 9)	

and support their child (Brown et al. 2020). The impact of parental provided distraction and communication has also been shown improve pain outcomes during procedures (McCarthy et al. 2010). Parental anxiety within this cohort scored a mean of 5.4 (SD 3.0). Parents were noted through their feedback to suggest the use of Smileyscope VR reduced parental stress, the ability to hear what was happening and converse with their child throughout also allowed them to encourage and support their child. The relationship between parental anxiety and patient outcomes is delicate (Brown et al. 2018, 2019b). The option and ability to engage with their child and support them throughout their procedure can not only reduce their anxiety but ultimately improve their child's outcomes.

The observed underestimation of pain in clinician-reported assessments, as opposed to self-reported or proxy-reported measures, underscores the inherent subjectivity of pain experiences. Clinicians, while relying on objective clinical indicators, may inadvertently overlook nuanced aspects of pain perception that are best captured through direct communication with patients or proxies. This disparity highlights the importance of incorporating patient and caregiver perspectives in comprehensive pain assessments to ensure a more holistic and accurate understanding of individuals' pain experiences (Berduszek et al. 2022; Choinière et al. 1990; Solomon 2001).

4.2 | Programmes and Device

Distraction device design and programmes have evolved to enhance feasibility in clinical setting (Armstrong et al. 2022; Xiang et al. 2021). The makeup and compatibility of VR devices, from basic phones to waterproof devices and computers with headsets and hand controls, significantly impact its success or failure, along with the staffing requirement and operational expertise (Hoffman et al. 2004; Markus et al. 2009; Mosso et al. 2009). Most devices within the literature feature a headset, often attached to a larger computer or equipped with hand controls. Bulky IT based equipment imposes a large burden on the clinical environment and staffing hindering easy implementation (Xiang et al. 2021). Some devices, like the one used by Hoffman et al. (2019), allowed flexibility for patients with facial burns, as the headset was placed near, not directly over the face (Hoffman et al. 2019). While this approach had advantages, Smileyscope device utilises a headset without hand controls or IT based equipment. This design facilitated inclusion for patients with hand burns, as 25% of our project participants had hand burns and could use the Smileyscope without handsets. Conforming to a wide variety of age groups, and sizes is paramount, and the adaptability of the VR device's fit is vital for broader and more effective adoption of Smileyscope in clinical settings. Our study showed that further development into the design and fit of Smileyscope to accommodate all age groups, should be considered due to majority of participants having concerns over the fit or weight of the device.

The appropriateness of VR programmes and concerns about potentially triggering memories of the initial burn injury, especially with fire-like scenarios, have been recurrent themes in VR literature. Additionally, the perceived cooling effect of snow or water may influence patients reported pain (Hoffman et al. 2020; Xiang et al. 2021). Previous studies show considerable variability in program content. Despite Smileyscope being mainly designed for needle procedures, its underwater diving element may play a part in the positive effect it had on our paediatric burn patients. While diverse programmes allowed choice, recurrent themes suggest current programmes are more suited a younger age group. Developing age- appropriate programmes proves challenging in a paediatric setting, as emphasised by both patients and parents. Ongoing development of programmes is a necessity for Smileyscope to be successful in this clinical environment.

Training and infection control also play a part on appropriateness of device within a clinical environment. Devices requiring lengthy staff training and cleaning procedures pose barriers to implementation (Markus et al. 2009). Given ongoing challenges of staffing and clinical workloads in hospitals, introducing new techniques or equipment requiring extensive training or dedicated staff is often not feasible. Thus, our chosen device needed to be easy to set up, require minimal training and not require dedicated staff. Smileyscope proved ideal, staff required no specific training, effortlessly navigating the menu within minutes, and it was extremely child friendly, allowing children to choose their own programmes. For infection control, efficient cleaning is crucial in a busy paediatric burn's outpatient department. Smileyscope surfaces were found to be easily wiped down by appropriate clinical wipes.

4.3 | Compliance

Increased compliance during procedures with the use of virtual reality has been mentioned within the literature (Hua et al. 2015; Tennant et al. 2021). While we were focusing on the acceptability and usability of VR within our unit, increased compliance was mentioned by many clinicians along with the reduced need for analgesia and extra clinical staff. This highlighted the ability of VR to perform painful procedures with less resources. However, there were suggestions that compliance was difficult in some patients with their need to control and view what occurred. Some suggest that compliance in younger children may be difficult due to their inability to voice pain or concern at an early age (Lauwens et al. 2020). The need for control over a procedure could be considered a child's response to a perceived threat or potentially painful event (Brown et al. 2018). This needs to be considered with an understanding from clinicians that VR may not be effective to all patients within the targeted age group.

4.4 | Strengths and Limitations

Nursing staff not only deal with distraught patients but within the paediatric setting this is also complicated by the presence of parents/caregivers. Burns procedures are painful, parents are protective of their child and children are anxious of what can occur. This can all lead to heightened procedural anxiety. The use of Smileyscope within our setting enabled nursing staff to complete painful procedures with minimal pain relief, provided distraction to both patients and parents that allowed nursing staff to complete these procedures within short time frame and with less resources. Nursing staff found the use of Smileyscope easy to integrate into the unit requiring minimal training.

We understand that the use of this survey can limit the depth of qualitative data obtained by participants. Grounding our study in the COM-B framework enabled the identification of determinants influencing the device's feasibility, providing a theoretical foundation for future models and studies testing immersive VR devices. While Smileyscope has been studied in other paediatric settings, its adaptability into the paediatric burn environment was unknown until now.

An acknowledged limitation is that all VR studies (including this one) in paediatric setting have focused on children aged 4 years or older, excluding significant cohort of burn patients (Smith, Wang, and Colloca 2022). A previous study conducted within our unit highlighted that over 60% of our cohort are under the age of four (Stockton, Harvey, and Kimble 2015), indicating the need for other distraction methods not addressed by VR.

This study did have its limitations. Due to the age of child participants, fewer than half of all children could provide feedback. Assent was obtained from children aged 8 years or over to provide feedback, limiting the amount of data able to be obtained. Thus, making it difficult to draw conclusions from the self-reporting child results. As there was no comparison between the intervention and standard treatment, we can draw no comparison to standard treatment. As an unfunded, clinician driven project carried out in a busy department with competing demands, completion and facilitation of research data collection was challenging. As a result, many patients were missed, and some feedback not obtained.

4.5 | Future Considerations

As Holbert et al. (2021) identified, first dressing changes outside of a dedicated burns unit can have a negative impact on a patient's pain and anxiety. While it is not feasible for all patients to be transferred to a specialist hospital for treatment, consideration into how we can improve outcomes for these patients within their local service is paramount. The experience, knowledge and access to appropriate technology required to understand and adequately treat the complexity of paediatric burns can play a large part, however, having access to devices to improve pain and anxiety that are easily implemented may be a simple start. Smileyscope is already being implemented into regional hospitals within our state to improve children's experience during insertion of needles. This project will help identify other procedures that Smileyscope can have a beneficial pain alleviation effect. Due to its relatively inexpensive, approximately $5000 AUD and easily implemented design, it is a viable option for regional and rural hospitals to have when performing painful paediatric procedures that may ultimately have a positive effect on long term healing and scar development.

There is a need to conduct further clinical studies within the paediatric burn environment A randomised control study with both control and intervention arm will provide a greater depth of data and provide further information into the use of Smileyscope in terms of pain and distress during paediatric burn dressings. In addition to an RCT, an exploratory qualitative study will provide further information of clinician attitudes, patient experience and caregivers perception on Smileyscope that quantitative studies may be unable to capture or expand on.

5 | Conclusion

This is the first study to explore the feasibility and acceptability of Smileyscope within a tertiary paediatric burns centre. While the device was originally designed to assist paediatric patients during needle procedures, it has shown promise within the paediatric burn's population. One concern is finding appropriate devices that fit within a clinical environment with appropriate immersive programmes, are easily implemented, require minimal training and setup, can be used by the majority of patients irrespective of location of burn or age and easily replicated into regional and rural hospitals. There is further opportunity to continue to explore the use of Smileyscope within the paediatric burn's environment, particularly with this being a relatively small cohort of patients. While the need for further development of programmes was prominent within our feedback, the device was well accepted from patients and parents or caregivers. Clinicians found the device easy to implement within the clinical setting requiring minimal training or setup.

Author Contributions

K.S., T.A.D., K.P. and B.G. contributed to the development and design of the study. K.S. prepared the manuscript. All authors provided review and edits of the draft manuscript and approval of the final manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Peer Review

The peer review history for this article is available at https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/jan.16417.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.